

CloudStack vs OpenStack

	CloudStack	OpenStack
Website	http://cloudstack.apache.org	http://openstack.org
Current version	4.0.1-incubating	Grizzly (2013.1)
Open source	Yes	Yes
Licence	Apache Licence 2	Apache Licence 2
Components	Management server	Nova (compute)
	Hypervisor	Swift (object storage)
	Storage nodes	Networking (Quantum)
	Zone, Pod,	Glance (image service)
	Cluster, Host,	Keystone (identity management)
Platform	Primary storage,	Horizon (GUI interface)
	Secondary storage	Block storage (Cinder)
Hypervisor support	Xen, KVM, VMware, Citrix XenServer, OVM, and Bare Metal	Xen, KVM, UML, LXC and VMware
Disk images	Yes	Yes
Console	Yes	Yes
User management	Yes	Yes
RightScale support	Yes	Yes
API	Yes	Yes
AWS API Compatibility	Yes	Yes
Live migration	Yes	Yes
VMware hypervisor support (ESX, ESXi, vSphere)	ESXi	Yes
Snapshots	Yes	Yes
Self-service user portal	Yes	Yes

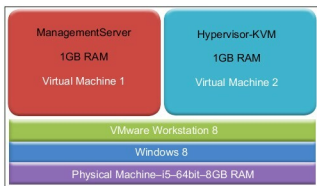


Figure 4: Simple deployment architecture of CloudStack

management server and another virtual machine to act as the cloud infrastructure—the host running the hypervisor software.

The management server manages cloud resources, and the administrator can manage and interact with the management server by using a UI and APIs. It also manages the assignment of guest VMs to particular hosts, the assignment of public and private IP addresses, templates and ISO images, as well as snapshots.

The CloudStack API

The CloudStack API supports three access roles: root admin, domain admin and user. The root admin can access all the

features in addition to both virtual and physical resource management; the domain admin can access only the virtual resources that belong to the administrator's domain, while the user can access the features that allow the management of the user's virtual machines, storage and network.

To use the CloudStack API, knowledge of Java or PHP, HTTP GET/POST and query strings, XML or JSON, URL of the CloudStack server, and API key and secret key is necessary. 

References

- [1] <http://en.wikipedia.org/wiki/CloudStack>
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- [4] <http://www.networkcomputing.com/cloud-computing/the-stack-wars-openstack-vs-cloudstack/240000933>
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- [6] <http://www.skidshare.net/OREillyOSCON/comparing-open-source-private-cloud-platforms>

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